# 90 Seconds: Linking video creators and clients with Google Cloud

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With Google Clou
Platform, 90 Sec
delivers reliable,
cloud video prod
paves the way to
video analysis wi
machine learning
technologies.

## Google Cloud Results

- Scales to support

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## About 90 Seconds

90 Seconds is a leading video creation platform that manages a marketplace of 12,000 video creative professionals in more than 160 countries across 70 categories. The company's technology empowers brands to plan, shoot, and edit video from anywhere in the world. 90 Seconds' 3,000-plus clients includes many global enterprise brands.

Industries: Other, Technology

growing demand for cloud video production
- Accelerates software development
- Captures and analyzes data from multiple services to facilitate decision making

creati
profes
and 3K brands

Location: Singapore

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Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)

Compute Engine (https://cloud.google.com/compute/)

Cloud SQL (https://cloud.google.com/sql/)

BigQuery (https://cloud.google.com/bigquery/)

Cloud AutoML (https://cloud.google.com/automl/)

Founded in 2010 in New Zealand, 90 Seconds (https://90seconds.com/) is a video creation platform that enables brands to create videos anywhere in the world. "On the one side, we have brands that want to create videos, and on the other side, we have creatives such as directors, videographers, editors, animators, and producers," explains Dat Le, Director of Data Science and Engineering at 90 Seconds. "We also provide automated workflow tools to make the video creation and delivery process as easy as possible."

A few years ago, 90 Seconds received millions of dollars in venture capital, and its founders began using new technologies to scale the business. Another, more recent injection of funds gave 90 Seconds further impetus to grow, and the business now has about 160 team members worldwide – 60 of whom are based at its headquarters, now in Singapore.

For Le, operating in the video production sector is a challenge and opportunity for 90 Seconds. "It's a very complex process and requires deep expertise from the people involved," he explains. "However, this also makes entering the market difficult, so we don't see too many competitors."

Despite what Le describes as 90 Seconds' low profile, the business is continuing to grow rapidly. Multinational online services present innovative marketing and advertising opportunities and brands are keen to produce compelling video content.

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*—Dat Le, Director of Data Science and Engineering, 90 Seconds*

## Scalability, reliability, and performance

90 Seconds started operations in a colocated data center in the United States, but its rapid expansion saw the business run into scalability and cost issues – particularly around content delivery and storage. 90 Seconds then elected to move to another cloud service. Earlier this year, the business decided to migrate again, this time to Google Cloud Platform.

"Our engineering teams really drove the initial decision to move to Google Cloud Platform because of the scalability, performance, and reliability of the platform," explains Le. "When I joined the business, we began moving our entire data infrastructure to Google Cloud Platform."

Google's proven expertise in HD video delivery and analysis played a key role in 90 Seconds' decision to migrate to Google Cloud services. "For example, we could help creators upload large video files quickly to a service like YouTube, which is owned by Google," Le says. "Overall, we could consolidate everything onto Google to improve manageability and performance."

In less than a year, 90 Seconds was all-in on Google Cloud Platform – barring eight years' worth of HD video the business is exploring options to move into the platform from its original colocated data center.

90 Seconds has recorded a range of benefits from its use of Google Cloud Platform. "Controlling access to Google Cloud Platform services is extremely easy," says Le. "Single sign-on gives users role-based access to all Google services. I do not, for example, have to manage database access by creating and managing a specific database administrator account, with all the associated concerns about password sharing or password storage."

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*—Dat Le, Director of Data Science and Engineering, 90 Seconds*

## Faster software releases

90 Seconds has also deployed Kubernetes Engine (https://cloud.google.com/kubernetes-engine/) to run its applications in containers, providing a foundation to accelerate software development and deployment. "We wanted to build, test, and release software faster and Kubernetes Engine supported this approach," explains Le. "It was also a natural choice as there was a big community of people knowledgeable in the product, making it easy for us to recruit skilled engineers."

The simplicity and intuitive nature of the product made deployment easy. "We assigned the project to a junior data engineer, who did not know much about Kubernetes or how it operated," says Le. "However, he took on the challenge, got involved with the community, read some FAQs, studied some documentation, and completed the project in just two and a half weeks."

90 Seconds is also using BigQuery (https://cloud.google.com/bigquery/) as a warehouse for transactional, marketing, and finance data – ranging from tracking data about Google, Facebook, and YouTube applications to payment details for freelancers and service interactions with customers. The business then uses visualization tools to review various indicators and make decisions to improve products, seize opportunities, and operate more efficiently.

Le regards BigQuery as vastly stronger than competing solutions. "With BigQuery, we did not have to set up any big data systems or environments ourselves, while the pricing model supported the scaling of our business," he says. "We didn't have to spend an engineer's time determining how many virtual machine instances we needed to support increased demand – we just had to pay more as our use of the service rose, which was very simple for us."

90 Seconds also uses Compute Engine (https://cloud.google.com/compute/) to access infrastructure-as-a-service and Cloud SQL (https://cloud.google.com/sql/) to run its relational database.

"Google Cloud Platform has

played a key role in helping our business grow to this point. We see technologies like Cloud Vision API, Cloud Video Intelligence, and Cloud AutoML helping us become a more intelligent, valuable provider to brands in future."

*—Dat Le, Director of Data Science and Engineering, 90 Seconds*

## 23,000 videos

With Google Cloud Platform, 90 Seconds has grown to support a marketplace of 12,000 video creative professionals in more than 160 countries across 70 categories. The business has worked with more than 3,000 brands – including some of the world's most recognized organizations – to deliver high-quality video content. Creative professionals and brands have delivered more than 23,000 videos via 90 Seconds.

## Machine learning as next priority

With its Google Cloud Platform architecture now in place, 90 Seconds is evaluating the potential of machine learning through Cloud AutoML (https://cloud.google.com/automl/) to provide deeper and more relevant analysis for brands. "Services such as Cloud AutoML allow us to easily train machine learning models for video analysis," says Le. "Extracting content from videos – for example, footage of sunsets or of people – and analyzing how they contribute to performance in attracting viewers and engagement would help bring our business to the next level.

"Another area machine learning can really help us is in extracting data from customer briefs," he adds. "Usually these are supplied in free-text form and it takes us time to understand and define customer requirements. Machine learning will allow us to understand these requirements much faster."

Finally, the business plans to use machine learning to power a chatbot that will initially help give customers answers to frequently asked questions before moving on to more detailed conversations about new services and price points.

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Intelligence (https://cloud.google.com/video-intelligence/)
, and Cloud AutoML helping us become a more intelligent, valuable provider to brands in future," concludes Le.